Find the Mistake: Indexes, Ratios and Convergence Answer Key

Algebra 2

Quick Answers

1. (a) = 23, (b) = —
2. 24
3. (a) = 3, (b) = 1215
4. (a) = $x^4 + 8x^3 + 24x^2 + 32x + 16$, (b) = 24
5. (a) = —, (b) = 6
6. 1010
7. 1080
8. (a) = 2, (b) = 5, (c) = 315

What is verified

12 of 14 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 5. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSA-APR.C.5 – *problems 4, 7*
HSF-BF.A.2 – *problems 1, 2, 3, 5, 6, 8*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

1. *If they got 19: started the index at zero, so the fifth term was computed as if n were 4*
2. *If they got 25: started the sum at k = 0, adding one extra term to the four the notation asks for*
3. *If they got 10: subtracted consecutive terms instead of dividing, treating a geometric sequence as arithmetic*
4. *If they got 32: counted the third term as the one with index k = 3, which is really the fourth term*
5. *If they got −3: used the reciprocal of the common ratio, applying the formula with 2 in place of one half*
6. *If they got 1008: summed 21 terms by starting the index at n = 0 instead of n = 1*
7. *If they got 540: left the coefficient 2 unsquared, using 2 where the term needs 2 squared*

1. $a_n = 4n + 3$ for $n \geq 1$; a student gave 19 as the fifth term. (a) Correct fifth term. (b) The error.

Solution (a): The rule starts at $n = 1$, so the fifth term is the value at $n = 5$.

$$a_5 = 4(5) + 3 = 20 + 3$$

$$\boxed{23}$$

Solution (b) — instructor-judged. The student used $n = 4$: $4(4) + 3 = 19$. That is the *fourth* term. Because the rule is stated for $n \geq 1$, the first term sits at $n = 1$, so counting from zero shifts every term one place. Full credit names the starting index and says which value of n the fifth term uses. "They used the wrong number" does not identify the off-by-one.

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2. Evaluate $\sum_{k=1}^{4}(2k + 1)$; a student wrote 25.
Solution: The notation says k runs $1, 2, 3, 4$ — four terms, and no $k = 0$ term.

$$(2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + (2 \cdot 4 + 1) = 3 + 5 + 7 + 9$$

The correct total is $\boxed{24}$

The student's 25 is this total plus the $k = 0$ term, which is 1.

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3. 5, 15, 45; a student reported the ratio as 10. (a) Correct ratio. (b) Sixth term.
Solution (a): A common ratio is found by *dividing* consecutive terms, not subtracting them.

$$\frac{15}{5} = 3, \qquad \frac{45}{15} = 3$$

Both quotients agree, confirming the sequence is geometric. $\boxed{3}$

Solution (b): The sixth term is five ratio steps past the first.

$$a_6 = 5 \cdot 3^5 = 5 \cdot 243$$

$$\boxed{1215}$$

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4. $(x + 2)^4$. (a) Expand. (b) Correct coefficient of the third term (a student said 32).
Solution (a): Row 4 of Pascal's triangle is $1, 4, 6, 4, 1$, and the powers of 2 climb as the powers of x fall.

$$(x + 2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16$$

$$\boxed{x^4 + 8x^3 + 24x^2 + 32x + 16}$$

Solution (b): In descending powers, the third term is the x^2 term, which uses $k = 2$, not $k = 3$:

$$\binom{4}{2} \cdot 2^2 = 6 \cdot 4$$

so the coefficient is $\boxed{24}$

Using $k = 3$ gives $\binom{4}{3} \cdot 2^3 = 32$, which is the *fourth* term — the same off-by-one as problem 1, now inside an expansion.

5. A student wrote $\frac{3}{1-2} = -3$ for $3 + 6 + 12 + \cdots$. (a) Why it is meaningless. (b) Sum of $3 + \frac{3}{2} + \frac{3}{4} + \cdots$.

Solution (a) — instructor-judged. The infinite geometric sum formula is valid only when the common ratio lies strictly between -1 and 1. Here the ratio is 2, so the terms grow without bound and the series has no sum at all; the formula returns a number, but that number means nothing. A second, simpler check: every term is positive, so no sum of them could ever be negative. Full credit names the condition on the ratio; the sign observation is a welcome extra.

Solution (b): Here the ratio is $\frac{1}{2}$, which does satisfy the condition.

$$S = \frac{a}{1-r} = \frac{3}{1-\frac{1}{2}} = \frac{3}{\frac{1}{2}}$$

$\boxed{6}$

6. $a_n = 5n - 2$, $n \geq 1$. Sum of the first 20 terms; a student summed 21 terms and got 1008.

Solution: The first 20 terms are $n = 1$ through $n = 20$. Splitting the sum:

$$\begin{aligned}\sum_{n=1}^{20}(5n-2) &= 5\sum_{n=1}^{20} n \;-\; 2(20)\\ &= 5 \cdot \frac{20 \cdot 21}{2} - 40\\ &= 1050 - 40\end{aligned}$$

$\boxed{1010}$

The student's 1008 is 1010 plus the $n = 0$ term, which is -2: one term too many, taken from the wrong end.

7. Coefficient of x^2 in $(2x+3)^5$; a student got 540.

Solution: The x^2 term takes $(2x)$ twice and 3 three times:

$$\begin{aligned}\binom{5}{2}(2x)^2(2)^3 &= 10 \cdot 4x^2 \cdot 27\\ &= 1080x^2\end{aligned}$$

The whole factor $2x$ is squared, so the 2 is squared with it. The coefficient is $\boxed{1080}$

The student's 540 comes from writing 2 instead of $2^2 = 4$ — exactly half.

8. Positive geometric sequence with third term 20 and fifth term 80. (a) Ratio. (b) First term. (c) Sum of six terms.

Solution (a): Two ratio steps separate the third term from the fifth, so

$$r^2 = \frac{80}{20} = 4$$

and, since every term is positive, r is the positive root. $\boxed{2}$

Solution (b): The third term is two steps past the first, so $a_1 r^2 = 20$:

$$a_1 = \frac{20}{2^2} = \frac{20}{4}$$

$\boxed{5}$

Solution (c): With $a_1 = 5$, $r = 2$ and six terms,

$$S_6 = \frac{5(2^6 - 1)}{2 - 1} = 5(64 - 1) = 5(63)$$

$\boxed{315}$

A quick sanity check: $5 + 10 + 20 + 40 + 80 + 160 = 315$.